Ziyu (Victor) Zhao

ziyuzhao2027@163.com | +86-184-3947-1396 | github.com/12sqawdwq | 12sqawdwq.github.io

Interests: computer architecture & AI hardware, HW/SW co-design, high-performance computing systems, embedded systems

Education

Xi'an Jiaotong-Liverpool University (XJTLU) & Univ. of Liverpool (dual degree) Sep 2023 – Jul 2027 (exp.)

BEng Microelectronic Science and Engineering with Contemporary Entrepreneurialism

- **UK-classified average (University of Liverpool): Year 1: 73/100, Year 2: 78/100 — both First Class**. **University Academic Achievement Award**, 2025/26.
- Core modules: Microprocessor Systems (94), Electromagnetism & Electromechanics (90), C/C++ Programming (87), CMOS Digital Integrated Circuits (86), Electrical Circuits (86), Analogue & Digital Electronic Circuits (86), Digital Electronics II (84).
- Current (2026/27): Quantum Physics; Micro/Nano Integrated Circuits; Final Year Project (multi-channel electrochemical sensing platform).

Publications & Patents

- **Z. Zhao**, "PAD-Former: A Two-Stage PI-RADS Classification Framework with Multi-Modal Anomaly Diffusion and Cross-Attention," *IEEE BIBM 2025*, pp. 6162–6169, doi: 10.1109/BIBM66473.2025.11356522. [**First author; CCF-B**]
- **Z. Zhao**, "ReFER: Learned Soft-Region Priors for Clinically Significant Prostate Cancer Detection and Diagnosis in Biparametric MRI," *IEEE BIBM 2026*. [**First author; CCF-B; accepted, to appear**]
- "EDRF: Evidence-Decoupled Reliability Fusion for Case-Level csPCa Risk Assessment, *IEEE BIBM 2026*. [**Second author; CCF-B; accepted, to appear**]
- "Clinically Significant Prostate Cancer Detection at Biparametric MRI with Residual-Guided Deep Learning: Multicenter Development, External Evaluation, and Reader-Support Analysis," *Radiology: Artificial Intelligence* [**Under review; 2025 JCR IF 20.1, Q1 (ranked 1/217 in Radiology & Medical Imaging); CAS Zone 1 Top**]
- Finite-element modeling of eyelid–eyeball biomechanics for transpalpebral intraocular-pressure measurement. [**Journal manuscript in submission**]
- **3 Chinese invention patent applications** (applicant: LanZhi Medical; technical solutions and specifications drafted by me): two on probe-advancement control algorithms for tonometry, one on an alignment-aperture calibration method for probe positioning.

Research Experience

AI for Prostate MRI: csPCa Detection, Segmentation & Multimodal Fusion 2024 – Present

XJTLU Academy of Artificial Intelligence (AIAT, formerly AIAC) | Advisor: Assoc. Prof. Hong-Seng Gan

- Proposed **ReFER**, a region-referenced detector–diagnoser that learns soft anatomical priors from T2W and compares lesions against robust (Cauchy) regional references; with only **12.14 M parameters** it reached **case AUROC 0.843 / lesion AP 0.440** on PI-CAI (800 patients, patient-grouped 5-fold CV)—best case-level result among compared methods and second in lesion AP only to MedNeXt-L (∼5× larger).
- Raised small-lesion sensitivity at 0.5 false positives per case from **0.228 to 0.317**; designed PAD-Former (diffusion-based anomaly localization + cross-attention ViT) for 5-class PI-RADS grading; co-developed EDRF, an uncertainty-gated MRI–clinical fusion model reaching **AUROC 0.915 / AUPRC 0.792** on a 700-exam external test set (+0.034 / +0.082 over average late fusion).
- Ran an equal-data benchmark of nnU-Net v2 ResEnc, MedNeXt and SwinUNETR on a Slurm HPC cluster with locked external test cohorts, paired statistical testing and pre-registered decision gates.
- **Maintain GPU infrastructure for the research group: 2 racks, 16× NVIDIA A800**—container management, environment deployment and scheduling; built an open-source multi-server monitor (pixel_probetop).

Final Year Project: Multi-Channel Miniaturized Wireless Electrochemical Platform Sep 2026 – Present

XJTLU | based on the open-source NanoStat (ESP32 + TI LMP91000 analog front end)

- Extending a single-channel potentiostat to **up to 4 simultaneously sampled channels**, improving sampling conditions and noise budget, and rebuilding the host-side interaction logic with a modern web UI.
- Reconstructed the full board in Altium Designer: 42 components, 4-layer 39×20 mm PCB, datasheet-compliance review, Gerber/BOM/pick-and-place outputs.

ROS2 Autonomous Vehicle Control System Jun 2024 – Aug 2024

Fudan University, Intelligent Perception & Autonomous Systems Lab | Supervisor: Dr. Ruijiao Li

- Built LiDAR perception, SLAM localization, A* planning and PID speed/steering control on NVIDIA Jetson NX, integrated into one ROS2 navigation pipeline.

Medical Electronics: Electrical Stimulation & Pressure-Sensing Systems Nov 2023 – May 2024

East China Normal University, Intelligent Ultrasound Research Center | Supervisor: Jiangang Chen

- Core developer of a multi-channel electrical-stimulation system: STM32F103/F407 and ESP32 firmware, I^2C/CAN/Bluetooth integration, Flutter control app, and bio-impedance closed-loop output control.
- Built an STM32 pressure-array cushion with KNN-based recognition of abnormal sitting postures and MATLAB real-time visualization; earlier built a wearable ECG/SpO_2 monitor (MAX30102/GH3220) at the Shanghai Key Lab of Multi-Dimensional Information Processing.

Industry & Entrepreneurship

Shenzhen InnoX Academy – LanZhi Medical Jul 2025 – Present
Embedded & Algorithm Development Intern | home-use, eyelid-closed tonometer

- Developed Bluetooth communication and closed-loop control of the probe-advancement motor; then moved to algorithm R&D.
- Built eyelid–eyeball finite-element models in ANSYS and ABAQUS to analyse non-ideal measurement (eccentric probe advancement, inter-subject tissue variance, operator error), with mesh-sensitivity and parameter-identifiability checks.
- Turned simulation results into probe-advancement control, error-compensation, sampling and alignment-calibration strategies: **error within ±2.5 mmHg of the gold standard**; **ocular-hypertension detection rate raised from ∼60% (typical in industry) to 85%**; basis of the 3 patent applications.
- Project won **Second Prize, National Medical Insurance Innovation Competition** ("LanZhi Home Eyelid-Closed IOP Testing Service").

Finkey – Open-Source Hardware Security Key | Shuangfu Technology (Suzhou) Co., Ltd. 2025 – Present
Co-founder (company registered Aug 2025) & Embedded Software Lead

- Implemented OTP/HOTP/TOTP and **FIDO2** authentication in **Rust on Tock OS** on the Nordic nRF52840, building on Google OpenSK; re-partitioned MCU resources to cut firmware overhead and speed up the crypto path.
- Shipped **100+ units** in small batches to developers; showcased at hackathons and tech forums; incubated at XJTLU's entrepreneurship hub.

ByteDance – Volcano Engine, Data Department Dec 2025 – May 2026
Technical Operations Intern

- Benchmarked large models across reasoning, coding, tool use and instruction following; analysed LLM-agent products and AI-coding workflows (tool-chain structure, developer flow, limitations); triaged community issues into actionable reports for product and engineering teams.

DeepBlue Technology (Shanghai) Apr 2024 – Jun 2025
Embedded Software Engineer Intern

- Wheeled-biped robot: gait-control algorithms and motor-control interfaces; master–slave architecture with Raspberry Pi 5 (path planning, SLAM, depth-camera obstacle avoidance) and CAN-linked controllers using IMU fusion and PID stabilisation; gave public technical talks on DeepBlue AI and robotics products.

Sanyi 3D Technology (Qingdao) Jan 2024 – Mar 2024
Embedded Full-Stack Engineer Intern

- Designed and assembled large-format FDM printers; simulated delta-printer kinematics (stepper + belt drive) in SolidWorks; deployed Marlin firmware with PID motor and sensor control.

Selected Projects

Embodied AI & Robotic Systems

"Digital Fruit-Fly Brain": Connectome-Driven Control of a ROS Robot Sep 2026 – Present
RTX 5090D inference server + DIABLO (Xingtian) wheel-legged robot

- Built a whole-brain leaky integrate-and-fire network from the **FlyWire v783 connectome (138,639 neurons, 15.09 M connection records)**; RGB video drives visual-neuron proxies (HS, LC9) and fixed descending readouts are mapped to ROS velocity commands.
- Rewrote spike propagation as GPU event-driven updates captured in **CUDA Graphs**: 100 ms of brain time dropped from ∼223 ms to **46–48 ms (∼4.7×)** with identical whole-brain spike counts; validated on 16 episodes / 1,440 windows of continuous real video.

Xingtian LiDAR SLAM & Autonomous Navigation Sep 2026 – Oct 2026
Raspberry Pi 4B (ROS 2 Foxy) + Unitree L1 LiDAR/IMU + RTX 5090D compute server

- Designed a distributed stack: the robot only streams sensors and executes commands that expire after 0.3 s; the server runs KISS-ICP LiDAR odometry with IMU deskew, relocalization on saved maps and 10 Hz navigation; a Three.js web console dispatches tasks.
- Diagnosed path-tracking failures from 47 in-place turns (controlled point 0.56 m ahead of the turning centre) and rebuilt pursuit control on medial-axis routes; added link-loss return along the driven path and narrow-space recovery; autonomous explore-and-return ended within **7 cm** of the start.

OpenMV High-Speed Target Detection for RoboMaster Dart Guidance | GitHub Jun 2025 – Aug 2025

- STM32H7 + OpenMV multi-colour detection at **120–200 FPS** in dynamic scenes, with calibration-based visual ranging and a high-speed SPI link to the guidance controller.

Systems & Developer Tools

code-with-pi: Shared Persistent Terminal for Coding Agents | GitHub Aug 2026 – Present

- C11 broker that owns a ConPTY (Windows) or POSIX PTY (Linux) so an AI coding agent and its user observe or take over the same live shell; nonce-authenticated local IPC, read-only monitor mode, full on-disk transcript; CTest suites pass on Windows 11 and Linux.

pixel_probetop: Multi-Server GPU Monitor | GitHub Aug 2026

- Dependency-light Python hub and Linux agents with a web UI and Android viewer, used to watch the research group's A800 servers.

STM32 CLI Toolkit for Headless Embedded Development | GitHub Dec 2025

- Bash/Zsh wrappers around OpenOCD, CMake and ARM GCC: build-artifact detection, multi-series flashing, parallel builds and USB-reset recovery.

Embedded & Digital Design

STM32F446 Character-LCD Graphics Engine | **GitHub** Dec 2025

- Optimised 8-bit parallel HD44780 driver with graphics primitives, wireframe 3D rendering, particle effects and character-video playback on a 20×4 display (CMake + ARM GCC).

STM32 LCD1602 Real-Time Game | **GitHub** Nov 2025

- T-Rex runner using timer interrupts, EXTI button input, CGRAM sprite management and a non-blocking state machine (Microprocessor Systems, mark 94).

AXI-Stream Header-Insertion RTL Module | **GitHub** Jan 2025

- Verilog module with valid/ready handshaking, keep/last byte alignment, byte-count tracking and configurable data width, with a testbench.

AI & Engineering Software

ForgeSpec Studio: AI-Assisted Parametric CAD Generator | **GitHub** May 2026 – Aug 2026

- FastAPI service that turns natural-language requirements into a reviewable AssemblySpec and deterministic CadQuery source, exporting STEP/STL/Python with browser preview and a standards-backed parts library.

ANSYS_skill: CAE Workflow Validation | **GitHub** May 2026 – Aug 2026

- Gates automated ANSYS Mechanical runs by state (geometry, setup, solver input, solved, evaluated) and by result evidence; cantilever benchmark within **0.223%** of Euler–Bernoulli theory.

Sky Monitor API & Interactive Star-Map Wiki | **GitHub** May 2026

- FastAPI astronomy service using the HYG catalogue and Skyfield/JPL DE421 ephemerides to return visible targets for any GPS position and time, with a polar star-chart web demo.

Leadership & Outreach

- **President, XJTLU Science Innovation Association** (2023–2025, two terms, 200+ members): co-organised the **NVIDIA China Developer Day** at XJTLU (Oct 17–18, 2024; ∼200 attendees); led development of the X-SIA member/content platform (Cloudflare Workers, D1/R2, Expo); organised company visits to iFLYTEK, SenseTime and Baidu PaddlePaddle.
- **Campus Ambassador – Zhipu AI**: talks on LLM inference infrastructure (vLLM, SGLang, LangChain); launched a Dify campus partnership and hosted a Dify-based agent hackathon. **JetBrains**: on-site technical support at Apsara Conference 2024 & 2025. **MathWorks**: MATLAB/Simulink ecosystem and outreach at XJTLU.
- Regular technical speaker to audiences of 50–200 (NVIDIA Developer Day, DeepBlue forums, freshman taster lecture on Finkey, Sep 2026).

Honours & Awards

- **Second Prize**, National Medical Insurance Innovation Competition (LanZhi home tonometry service) [National] 2026
- **National 3rd Prize**, RoboMaster University Championship [National] 2024
- **1st Prize**, RoboMaster University Championship, Eastern Region [Regional] 2024
- **Successful Participant**, Mathematical Contest in Modeling (MCM, COMAP), Team 2601772 [International] 2026
- **Excellent Award**, China–U.S. Young Maker Competition [National] 2025
- **University Academic Achievement Award**, XJTLU [University] 2025/26

Skills

- **Languages & HDL:** C/C++, Rust, Python, Verilog, MATLAB, Bash, JavaScript/TypeScript
- **Hardware & Embedded:** STM32, ESP32, nRF52840, Tock OS, Raspberry Pi, Jetson, ROS2; CAN, SPI, I^2C, BLE; Altium Designer, Vivado, OpenOCD, CMake/ARM GCC
- **AI Infra:** multi-GPU cluster operations (16× A800: containers, environment deployment, scheduling), Slurm, CUDA Graphs (GPU event-driven acceleration), LLM serving frameworks (vLLM, SGLang), LLM benchmarking, agent/workflow platforms (Dify, LangChain)
- **Systems:** Linux, Docker, PTY/ConPTY, ROS 2, KISS-ICP, Cloudflare Workers
- **AI & Simulation:** PyTorch, nnU-Net, MONAI, OpenCV; ANSYS (APDL/Mechanical), ABAQUS, SolidWorks, CadQuery